01 — Executive Summary & Value Proposition

A verifiable trust asset, not another compliance spreadsheet



KEY TAKEAWAY

A working, deployed DPP platform — not a concept — that turns the EU's incoming ESPR/PPWR transparency mandate into a verifiable trust asset for NRW's polymer and chemical Mittelstand, built on cryptographic tamper-evidence and GS1-standard QR verification.

The problem

Europe's polymer value chain is being forced, in a short window, to prove things it has historically never had to prove: exactly how much recycled or bio-based content is in a given batch of resin, where that feedstock came from, what its carbon footprint is, and whether every claim on the label is independently verifiable rather than self-declared. The EU's Ecodesign for Sustainable Products Regulation (ESPR), the Packaging and Packaging Waste Regulation (PPWR, (EU) 2025/40), and the broader Digital Product Passport mandate turn "trust us" sustainability marketing into a hard compliance requirement with legal exposure attached — the EU's Green Claims direction explicitly targets unsubstantiated environmental claims.

For polymer compounders, recyclers, and converters — an industry segment heavily concentrated in NRW's Rhein-Ruhr and Rhineland chemical corridor — this creates an operational problem that spreadsheets and PDFs cannot solve at scale: mass-balance chain-of-custody (ISCC PLUS, REDcert2) has to be tracked batch-by-batch, certifications have to expire and get re-verified on a schedule, carbon footprint figures have to be distinguishable from marketing estimates, and every one of these facts has to be exposed through a QR-code-scannable, GS1-Digital-Link-compliant passport by the time ESPR's delegated acts start applying to plastics value chains.

Most mid-sized polymer processors — the Mittelstand backbone of NRW's chemical and plastics cluster — have neither the software engineering capacity nor the regulatory staff to build this themselves, and the generic ERP/PLM vendors serving this space were not built around DPP-specific data models.

What we have built

Polymer DPP is a working, deployed SaaS platform that gives polymer manufacturers, recyclers, and compounders a purpose-built Digital Product Passport system:

- **Industrial-grade material data capture** — ISO 1133 Melt Flow Index, density, tensile strength, PCR/PIR recycled content, bio-based content, and PEF/LCA-based carbon footprint, stored with exact decimal precision.
- **Multi-party trust workflow** — manufacturers submit batch data; independent, individually-credentialed third-party auditors formally verify or reject it before it is ever shown to a downstream buyer or consumer as fact.
- **Cryptographic immutability** — every batch record is SHA-256 sealed, and every state change is written to an append-only, hash-chained audit ledger with a live integrity-verification function.
- **Consumer- and auditor-facing transparency** — a GS1 Digital Link QR code on every batch resolves to a public passport page, plus dedicated auditor and buyer portals and a full ESG/compliance analytics suite.
- **Regulatory-adjacent analytics** — a computed Sustainability Readiness Index, PPWR 2030/2040 target tracking, supplier performance ranking, and illustrative EU plastic-tax exposure.

The value proposition

For manufacturers and recyclers: turn an unavoidable compliance burden into a competitive credential — a QR code any auditor can scan beats "we say our resin is 40% PCR."

For B2B buyers: a single point of truth for the sustainability data they must themselves report under their own CSRD/ESPR obligations, without re-keying supplier claims.

For NRW as a region: a homegrown, EU-compliant DPP infrastructure layer for one of Germany's largest chemical and plastics industrial clusters, reducing dependence on non-EU vendors for a function that becomes mandatory across the entire European polymer value chain within this decade.

Why now, why NRW

NRW is home to one of Europe's densest concentrations of chemical, plastics-processing, and recycling capacity, sitting directly alongside a world-class materials-science research base at RWTH Aachen. ESPR's delegated acts and PPWR's 2030

recycled-content thresholds create a hard deadline measured in a small number of years — the window to become NRW's default DPP infrastructure provider is open now.